

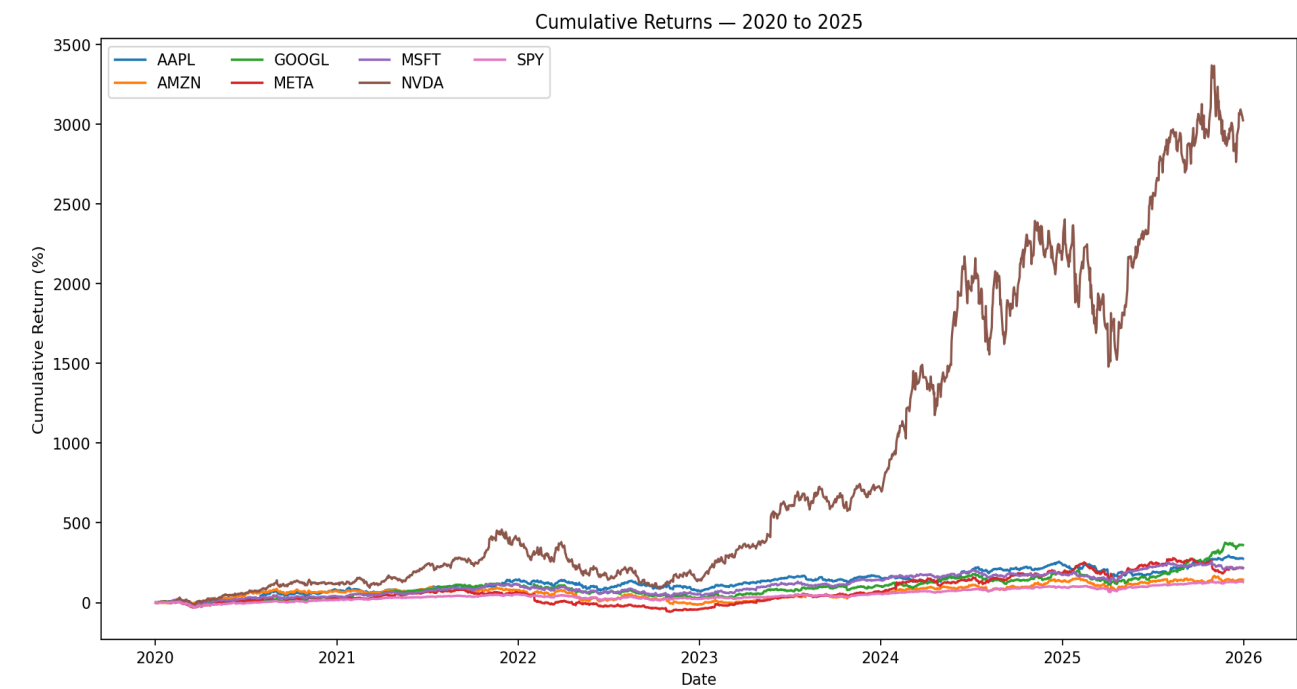
Stock Market Analytics & Time Series Forecasting

Python + SQL + Time Series + Power BI

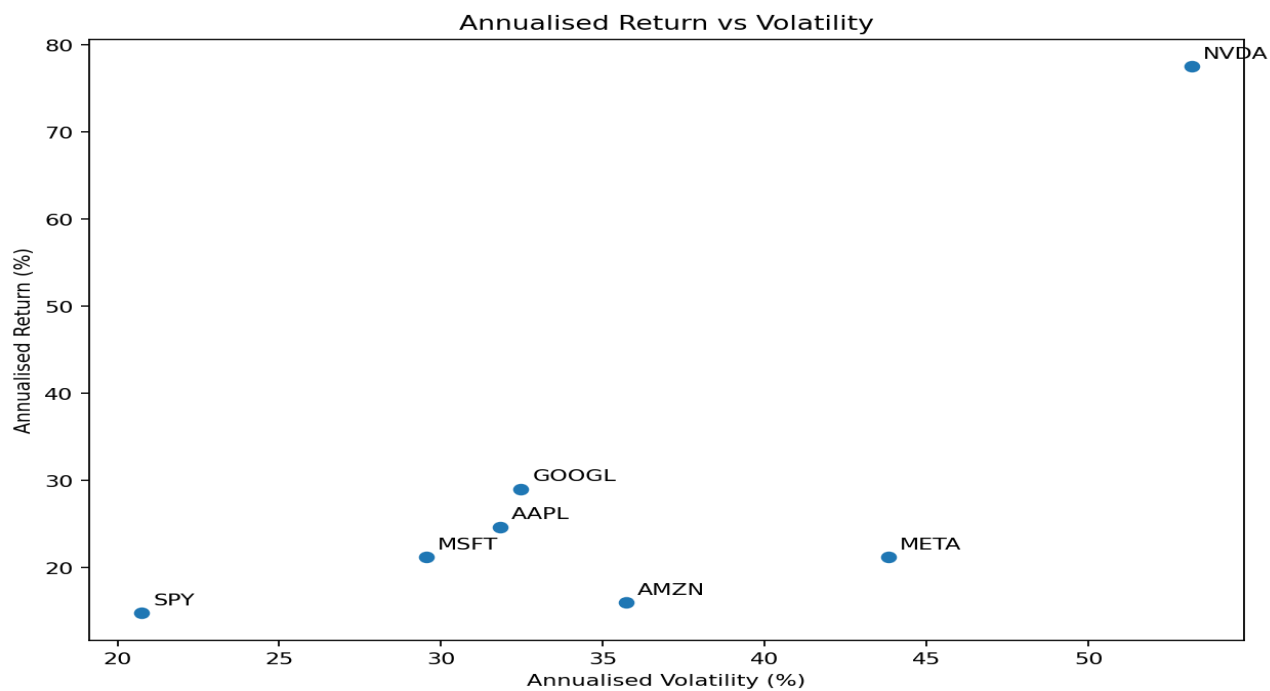
Executive Summary

This project analyses seven securities (AAPL, MSFT, NVDA, AMZN, GOOGL, META and SPY) over the fixed period 2 January 2020 to 31 December 2025. It combines financial analytics, SQL, risk metrics and time-series forecasting. NVDA generated the highest total return in the sample (3023.39%) and also the highest annualised volatility (53.18%). META experienced the deepest maximum drawdown (-76.74%). For SPY forecasting, the naive previous-close baseline outperformed ARIMA(5,1,0) on the held-out test period by RMSE (5.41 vs 11.09).

Metric	Result
Securities analysed	7
Trading days per ticker	1,508
Highest total return	NVDA - 3023.39%
Highest annualised volatility	NVDA - 53.18%
Deepest maximum drawdown	META - -76.74%
Highest Sharpe-style ratio (0% RF)	NVDA - 1.46
SPY total return	129.06%
Best test forecast	Naive Previous Close



Cumulative return comparison across the seven securities

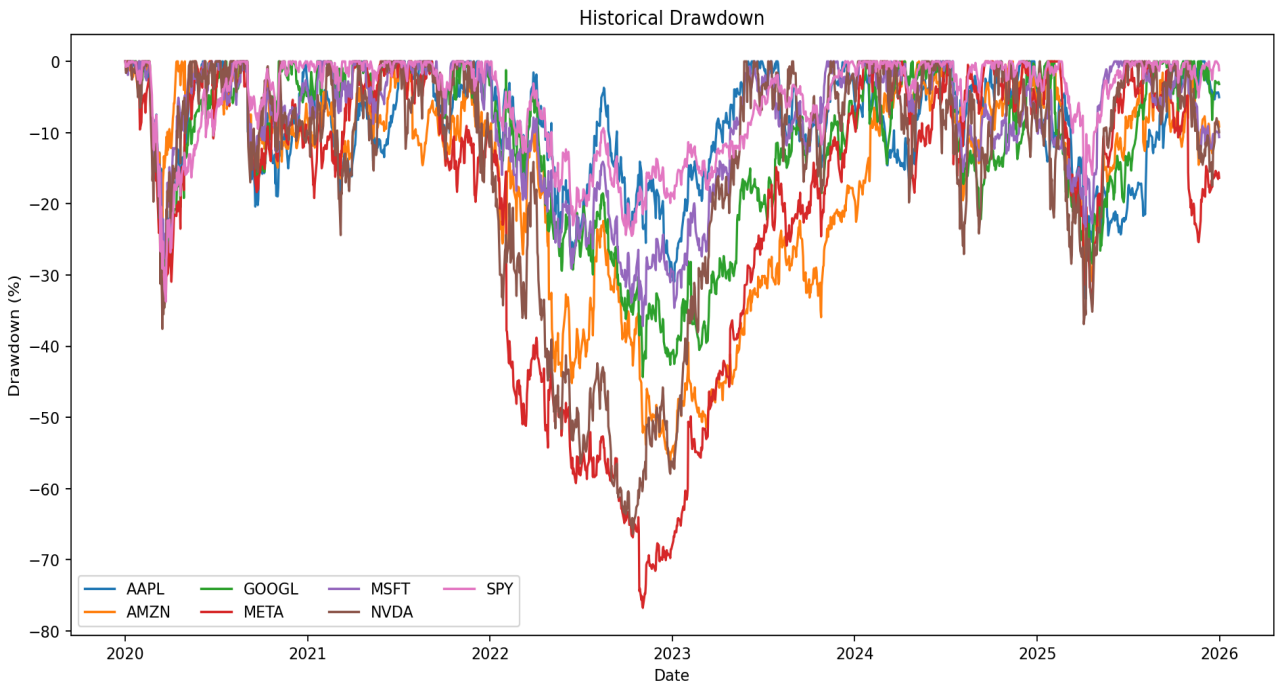


Annualised return versus annualised volatility

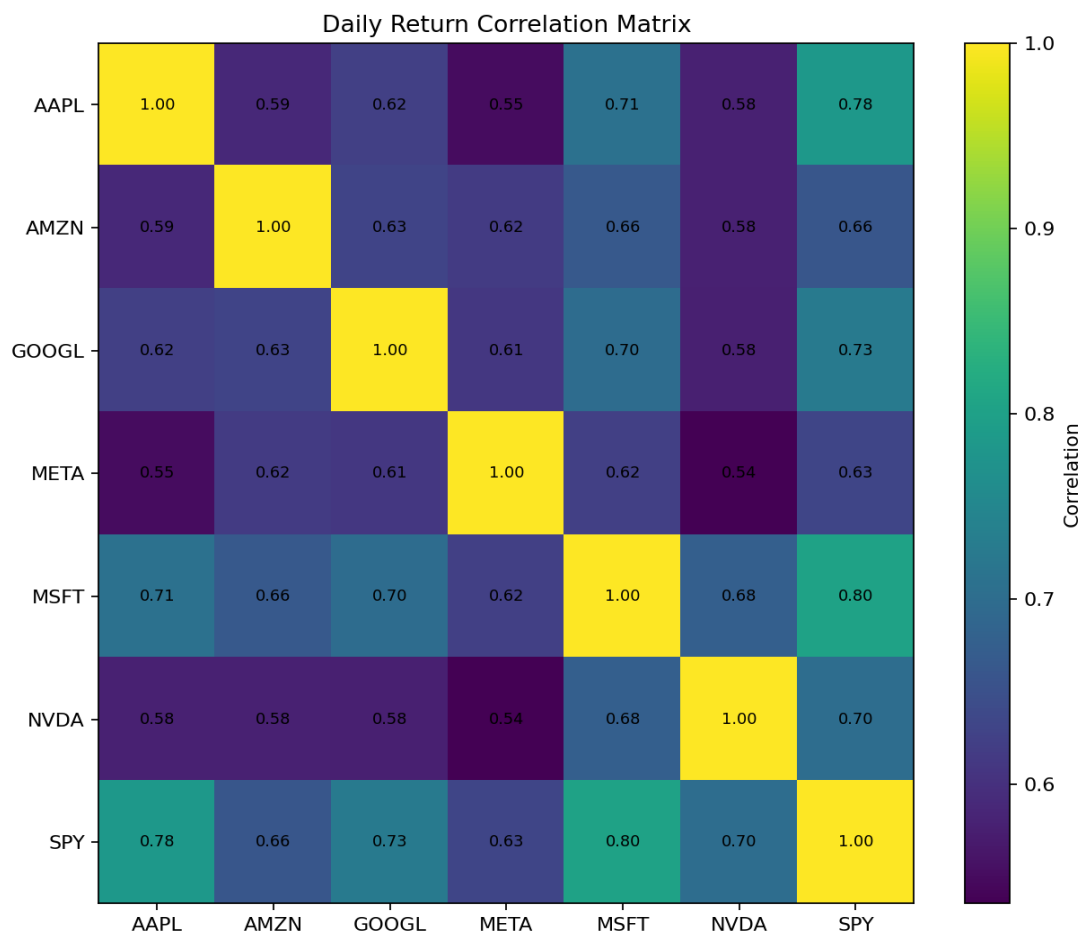
Risk & Drawdown Analysis

Return alone is not sufficient for comparing securities. The project therefore evaluates annualised volatility, drawdown, positive-day frequency and a Sharpe-style ratio under a stated 0% risk-free-rate assumption. NVDA leads the historical return ranking but also exhibits the highest annualised volatility. META shows the deepest peak-to-trough loss in the sample.

Ticker	Total Return	Ann. Return	Ann. Volatility	Max Drawdown	Sharpe-style
NVDA	3023.4%	77.5%	53.2%	-66.3%	1.46
GOOGL	360.8%	29.0%	32.5%	-44.3%	0.89
AAPL	275.1%	24.7%	31.8%	-33.4%	0.78
MSFT	217.1%	21.2%	29.5%	-37.1%	0.72
META	216.9%	21.2%	43.8%	-76.7%	0.48
AMZN	143.2%	16.0%	35.7%	-56.1%	0.45
SPY	129.1%	14.8%	20.7%	-33.7%	0.71



Historical drawdown by security

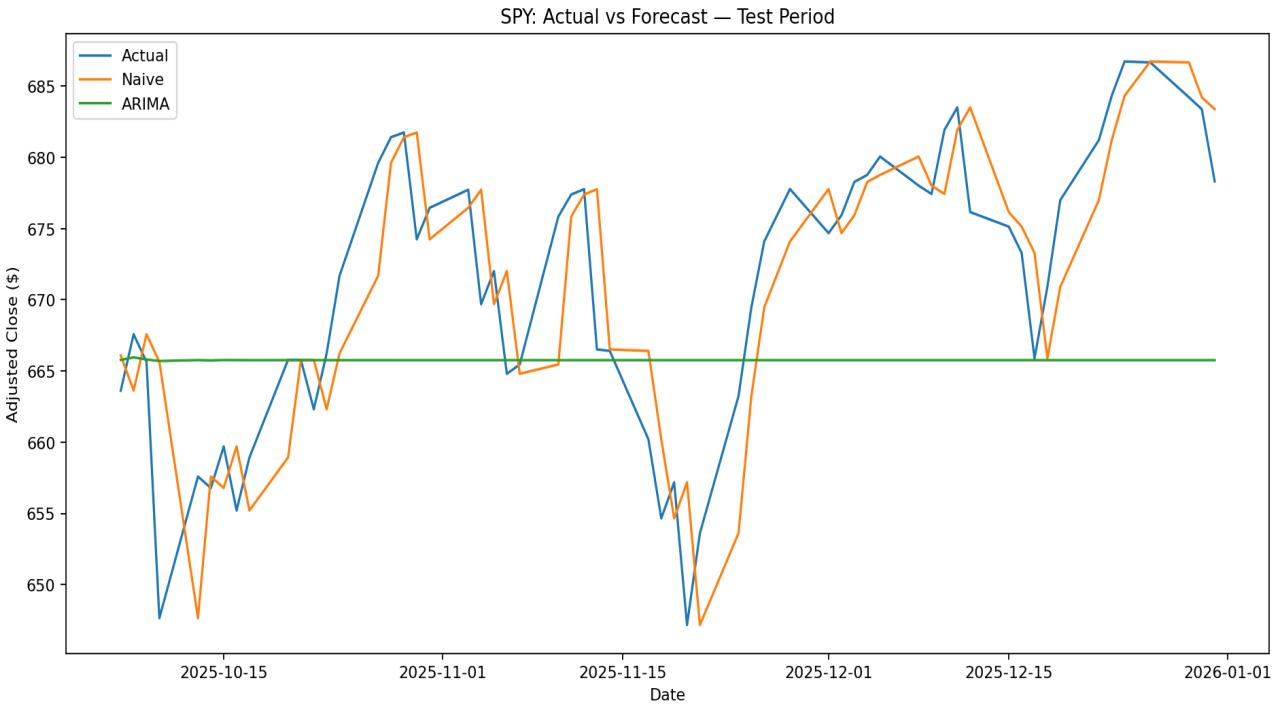


Daily-return correlation matrix

Forecasting Evaluation

SPY is used as the primary forecasting series. The final 60 historical trading observations are held out chronologically for testing. The models compared are a previous-close naive baseline and ARIMA(5,1,0). The naive baseline achieved lower error on all three reported metrics: MAE 4.16, RMSE 5.41, and MAPE 0.62%. ARIMA produced MAE 9.39, RMSE 11.09, and MAPE 1.39%. This illustrates why a simple baseline is essential in time-series forecasting.

Model	MAE	RMSE	MAPE
Naive Previous Close	\$4.16	\$5.41	0.62%
ARIMA(5,1,0)	\$9.39	\$11.09	1.39%



SPY actual versus naive and ARIMA forecasts

Conclusions & Portfolio Interpretation

The historical analysis shows large differences in return, volatility and drawdown across the selected securities. NVDA delivered the strongest historical return but with substantial volatility, while SPY showed lower historical volatility and a materially lower total return than the strongest individual technology stocks. Return correlations are positive but not identical, which is why return-based correlation analysis is more informative than comparing raw price levels.

The forecasting experiment provides an important modelling lesson: a more complex model does not automatically outperform a simple baseline. On the held-out SPY test period, the previous-close forecast was more accurate than ARIMA. The generated future 20-business-day ARIMA path should therefore be treated as a model illustration rather than a reliable prediction of future prices.

Key Recommendations

1. Evaluate historical return together with volatility and drawdown rather than ranking by return alone.
2. Keep a simple naive forecast as a mandatory benchmark for all future forecasting experiments.
3. Use chronological train/test splits and avoid leakage from future prices.
4. Extend future work with rolling-window backtesting, alternative ARIMA orders, exponential smoothing or lag-based ML.
5. Use the Power BI dashboard to distinguish historical facts, risk measures and model-generated forecasts.

Disclaimer

This is an educational analytics portfolio project. It is not financial advice. Historical returns and statistical forecasts do not guarantee future performance.